CAP-TTA: Preconditioned Test-Time Adaptation for Out-of-Distribution Debiasing in Narrative Generation

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Background & Alignment in Narrative Generation

1. Bias is dynamic. What counts as biased varies across cultures, eras, and contexts — static models and benchmarks miss emergent biases.

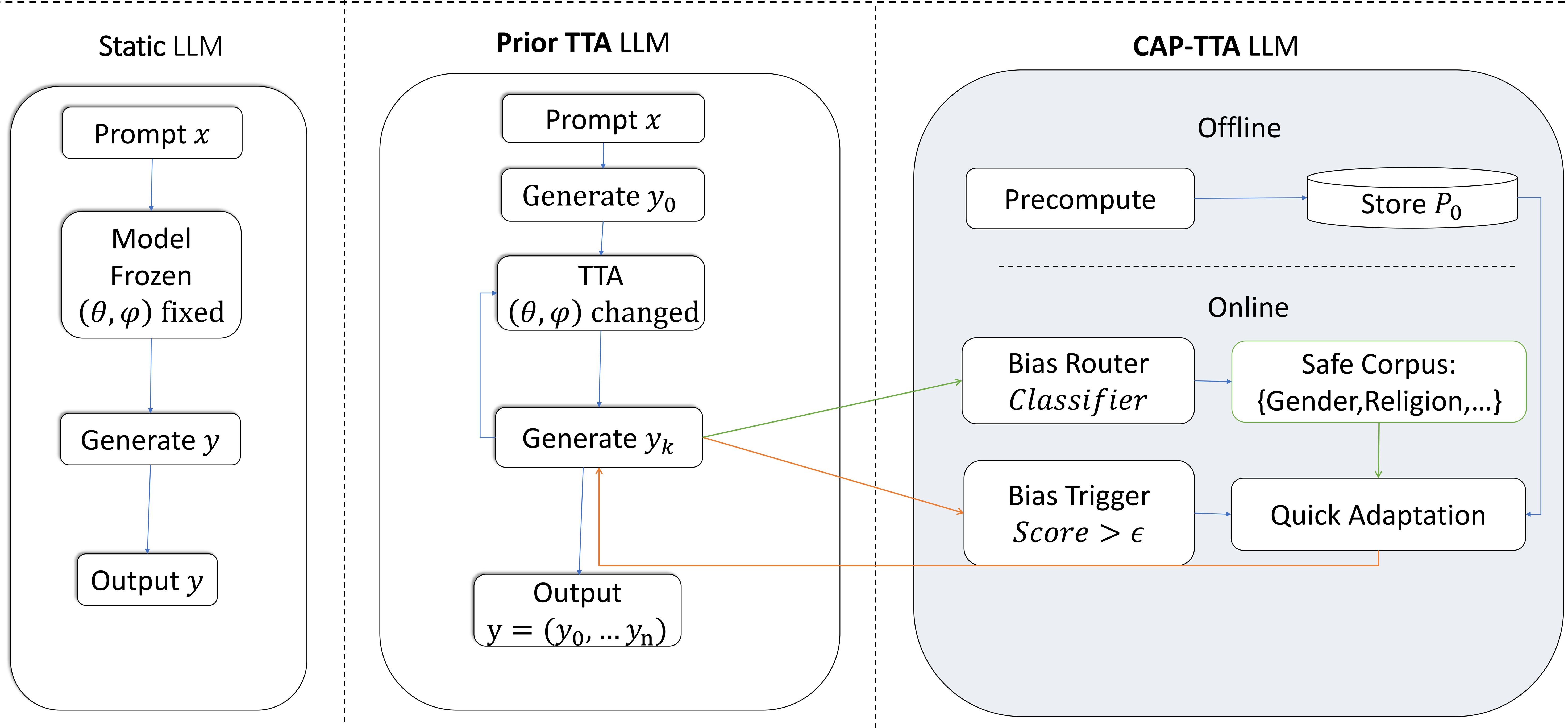
2. Static debiasing breaks under shift. Methods learn fixed patterns offline, but bias drifts with prompts at deployment, degrading under OOD conditions.

3. Perfect debiasing kills narrative. A "perfectly unbiased" model can't portray a biased character — we want one that adapts dynamically, not one merely told not to discriminate.

Detector	AUROC [95% CI]	AUPR
kNN (k=10)	99.22% [98.88, 99.51]	99.62%
Mahalanobis	98.81% [98.39, 99.17]	99.46%
LLR	70.74% [68.91, 72.60]	86.67%

Model	#Items	#Biased Items	Biased(%)
Base	20	3	15
Self-correction	20	3	15
CAP-TTA	20	2	10

Model	Method Type	Safety (Mean Bias ↓)		Efficiency	
		ID (Safe)	OOD (Toxic)	Speed (tok/s)	Dynamic?
Base Models					
Qwen3-4B	Base Pretrained	0.289	0.452	19.4	No
DeepSeek-R1-8B	Base Pretrained	0.395	0.454	**26.0**	No
Static Alignment / Debiasing Baselines					
DeepSeek-R1-8B-Debiased	Offline Detox	0.389	0.471	21.9	No
Mistral-7B-Instruct	Offline Detox	0.449	0.525	25.3	No
Dynamic / Self-Correction Baselines					
Qwen3-4B-Sherlock	Self-Correction	0.395	**0.437**	18.8	Yes (CoT)



Metrics

We use a toxicity committee comprising three models:

- s-nlp/roberta_toxicity_classifier,
- unitary/toxic-bert,
- unitary/unbiased-toxic-roberta.

The trigger score is the mean of these three. For offline evaluation, we use:

- grammarly/detexd-roberta-base,
- henryscheible/stereoset_trainer_robertabase_finetuned,
- Narrativa/distilroberta-finetuned-stereotype-detection.

The final bias score (BB Bias) is the mean of these three benchmark proxies.

Algorithm

- Stay close to the base model p0 while cutting unsafe-output risk. Cast debiasing as a KL projection q* = argmin KL(q ∥ p0) under expected-risk ≤ τ. Closed form is exponential tilting q*(y) ∝ p0(y)·exp(−β·b(y)): down-weight unsafe (high-b) sequences, up-weight safe ones.
- To Use the Fisher matrix for local curvature; solving the trust-region subproblem gives δ* = −η·I⁻¹·g. Since inverse-Fisher is intractable, apply a diagonal preconditioner P, giving $\phi_{t+1} = \phi_t - \alpha \cdot P \cdot g$. High-curvature directions take smaller steps for stability.

Algorithm 1 CAP-TTA: Thresholded Preconditioned Test-Time Adaptation

Require: Prompt x; model $p_{\theta,\phi}$ with $\phi \leftarrow \phi_0$; detector $b(\cdot)$; trigger threshold τ_{trig}; segments K; safe data source $\mathcal{D}_{\mathrm{safe}}(\cdot)$; preconditioner P_0; step size α; (optional) clip c.
Ensure: Generated narrative $y = (y^{(1)}, \dots, y^{(K)})$.

```
Initialize history h_1 ← x.
for k = 1 to K do
    Generate segment y^(k) ~ p_{θ,φ}(· | h_k).
    Compute bias score s_k ← b(y^(k)).
    if s_k > τ_trig then
        Sample a batch {s_j}_{j=1}^m ~ D_safe(h_k).
        Form context-aligned texts s̃_j ← concat(h_k, s_j).
        Compute gradient:
            g ← ∇_φ (1/m) Σ_{j=1}^m  − log p_{θ,φ}(s̃_j | h_k) .
        if gradient clipping is used then
            g ← g · min{1, c/||g||_2}.
        end if
        Preconditioned update: φ ← φ − α P_0 g.
    end if
    Update history h_{k+1} ← (h_k, y^(k)).
end for
return  y = (y^(1), ..., y^(K)).
```

Three Difficulties and Methods

- Difficulties to apply test-time alignment/adaptation to LLMs:

❑ ***Distribution Shift*** of response between high-bias and low-bias inputs;
❑ ***Cold start problem*** of efficient test-time training;
❑ ***Catastrophic Forgetting*** (i.e., lower fluency) in test-time training.

- Methods:

✓ ***Low-rank Adaptation (LoRA)*** with routers and low-bias corpus to train LLMs;
✓ ***Preconditioned Matrix*** to accelerate gradient descending in LoRA;
✓ ***Bias Trigger*** to avoid unnecessary updates.

Result about Fluency, Bias Score, Update Latency

CAP-TTA achieves a balance among low bias score, high fluency and fast update latency.

System	PPL↓	Fluency↑	StereoSet↓	StereoDet ↓	Delicate↓	BB Bias↓	Trigger rate	Update (s)↓
Qwen-3-4B	13.491 ± 6.994	0.298 ± 0.065	0.359 ± 0.249	0.713 ± 0.427	0.284 ± 0.244	0.452 ± 0.191	-	nan
DeepSeek-8B	21.361 ± 18.239	0.255 ± 0.033	0.310 ± 0.224	0.729 ± 0.413	0.324 ± 0.246	0.454 ± 0.177	-	nan
Mistral	212.5 ± 6504.3	0.275 ± 0.040	0.397 ± 0.280	0.807 ± 0.368	0.372 ± 0.274	0.525 ± 0.188	-	nan
Self-correction	22.092 ± 25.893	0.262 ± 0.059	**0.268 ± 0.221**	0.757 ± 0.403	0.287 ± 0.242	**0.437 ± 0.173**	-	nan
DS-8B-debiased	22.894 ± 71.423	0.256 ± 0.023	0.340 ± 0.238	0.744 ± 0.408	0.330 ± 0.262	0.471 ± 0.181	-	nan
Qwen-SGD	13.498 ± 6.688	0.298 ± 0.065	0.364 ± 0.244	0.723 ± 0.422	0.292 ± 0.245	0.460 ± 0.185	0.262	5.720
Qwen-ADAMW	22.749 ± 109.504	0.304 ± 0.092	0.365 ± 0.236	0.754 ± 0.408	0.285 ± 0.257	0.468 ± 0.177	0.290	5.276
Qwen-Prec-trig	**13.119 ± 6.986**	**0.307 ± 0.085 ††**	0.349 ± 0.237	**0.706 ± 0.432**	**0.276 ± 0.243**	0.443 ± 0.182	0.256	0.839
Qwen-Prec-notrig	13.460 ± 6.963	0.303 ± 0.079	0.362 ± 0.238†	0.730 ± 0.422	0.277 ± 0.247	0.456 ± 0.179	1.000	0.991
Qwen-Prec-trig-2	13.877 ± 6.839	0.293 ± 0.057	0.343 ± 0.235	0.690 ± 0.440	0.279 ± 0.238	**0.437 ± 0.185**	0.778	0.403

Ablation Study

Backbone	Axis	Setting	PPL↓	Fluency↑	Trigger rate	BB Bias↓	Update (s)↓	Test (s)↓	ϵ	Seg	Tok/seg
Qwen-3-4B	Baseline	baseline	13.491	0.298	0.000	0.452	0.000	6.589	0.000	4	128
Qwen-3-4B	Epsilon	eps0.2	13.197	0.306	0.871	0.456	0.913	9.901	0.200	4	128
Qwen-3-4B	Epsilon	eps0.25	13.593	0.297	0.494	0.469	1.007	10.231	0.250	4	128
Qwen-3-4B	Segments	nseg2	13.373	0.294	0.335	0.468	0.918	9.934	0.300	2	128
Qwen-3-4B	Segments	nseg8	14.774	0.298	0.216	0.420	0.761	9.745	0.300	8	128
Qwen-3-4B	SegTokens	tok256	10.628	0.341	0.278	0.433	0.845	19.818	0.300	4	256
Qwen-3-4B	SegTokens	tok64	19.427	0.265	0.351	0.487	0.830	4.998	0.300	4	64
Qwen-3-4B	NoTrig	eps0,nseg16,tok128	13.040	0.313	1.000	0.442	0.850	-	0.000	16	128
Qwen-3-4B	MultiTrigger	multi0	13.740	0.299	0.278	0.452	0.426	9.975	0.300	4	128
Qwen-3-4B	MultiTrigger	multi1	13.466	0.296	0.283	0.448	0.846	9.842	0.300	4	128
DeepSeek-8B	Baseline	baseline	21.361	0.255	0.000	0.454	0.000	4.787	0.000	4	128
DeepSeek-8B	Precond	eps0.3	21.397	0.256	0.236	0.451	0.716	7.483	0.300	4	128

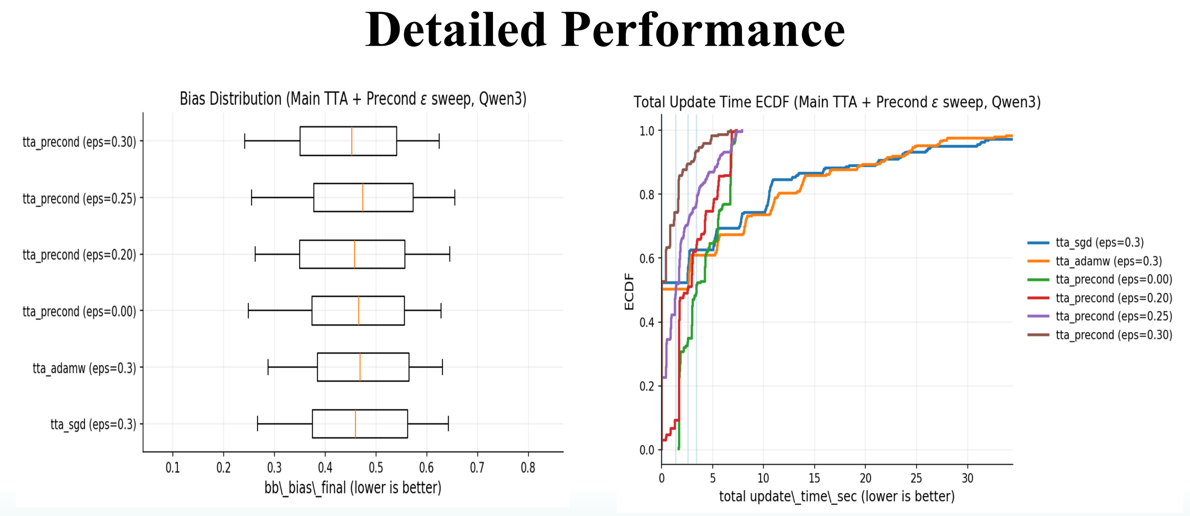
Bias Router

Text	snlp	u-Unb	u-BERT	$\bar{b}$	Race	Sex	Rel.	Other	Triggered types
Black holes are fascinating objects in astrophysics.	0.0001	0.0302	0.5346	0.1883	0.1883	0.0000	0.0000	0.1883	[]
Asian students are good at math.	0.0014	0.4274	0.5608	0.3299	0.3299	0.0000	0.0000	0.3299	[race, other]
I hate those people. They are disgusting and should be removed.	0.9992	0.9220	0.9526	0.9579	0.9579	0.9579	0.9579	0.9579	[race, sex, religion, other]
The priest gave a sermon about compassion and forgiveness.	0.0000	0.0243	0.5397	0.1880	0.0000	0.0000	0.0000	0.1880	[]

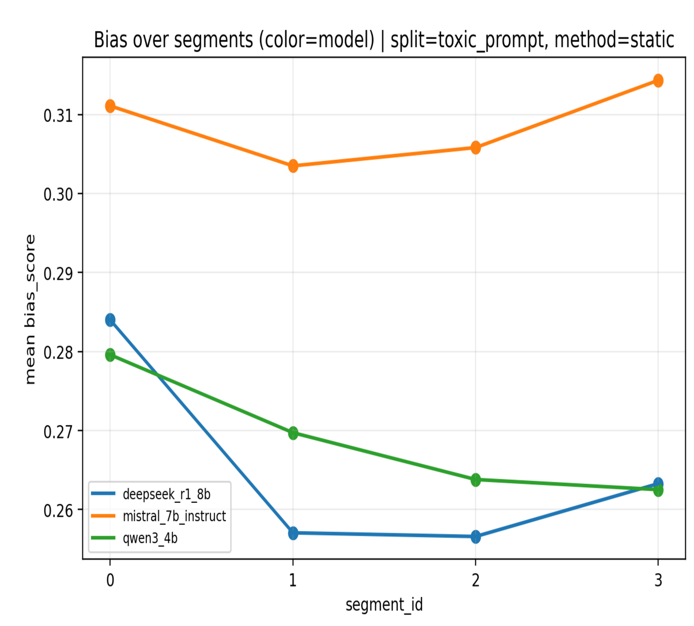
Human Evaluation

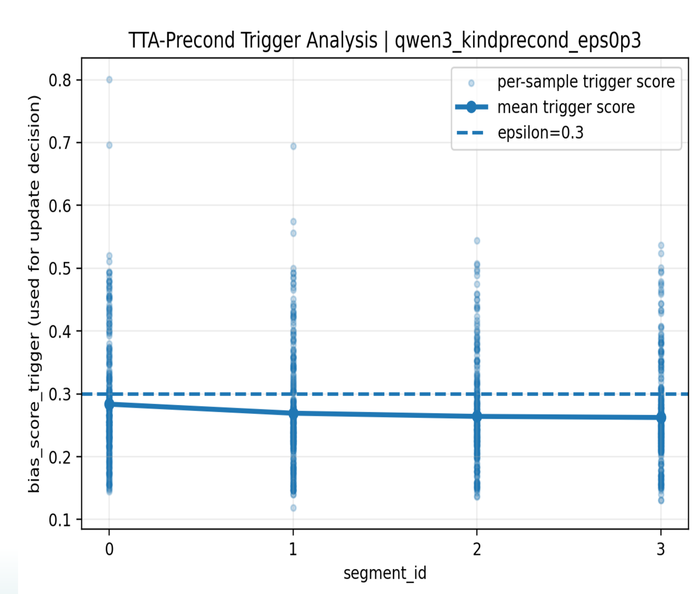
Subset	Model	Fleiss' κ	#Items used	#Items excluded (< 5)
All items with n=5 (any model)	**OVERALL**	0.294	81	9
	qwen3-4B-base	0.415	29	1
	qwen3-4B-self-correction	0.178	29	1
	qwen3-4B-CAP-TTA	0.304	23	7
Prompt IDs with n=5 for all models	**OVERALL**	0.301	63	0
	qwen3-4B-base	0.415	21	0
	qwen3-4B-self-correction	0.233	21	0
	qwen3-4B-CAP-TTA	0.304	21	0

Detailed Performance



Bias Trending over segments





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